

## PROBLEM SET 2

### ECON301—ECONOMETRICS I

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**Instructions:** Due on 25/2/2022; scan or type your work and submit it through the link in Moodle. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

1. Suppose that we have  $n$  independent observations  $(X_i, Y_i)$  of the random vector  $(X, Y)$ . Under Assumptions 1 to 4 find and justify the moments below. Try to make use as much as possible of the Theorems seen in the lectures.

- (a)  $E(\bar{Y}|X's)$
- (b)  $E(\hat{\beta}_0 + \hat{\beta}_1 X_1|X's)$
- (c)  $\text{Var}(\hat{\beta}_0 + \hat{\beta}_1 X_1|X's)$
- (d)  $\text{Var}(\bar{Y} - \hat{\beta}_1 \bar{X}|X's)$
- (e)  $\text{Cov}(\bar{Y}, \hat{\beta}_1|X's)$
- (f)  $E(\bar{Y} - \hat{\beta}_1 X_1|X's)$
- (g)  $\text{Var}(\bar{Y} - \hat{\beta}_1 X_1|X's)$

2. Under Assumptions 1 to 4 derive  $E(\hat{\beta}_0|X's)$ . *Hint: make use of  $E(\hat{\beta}_1|X's)$ .*
3. Under Assumptions 1 to 4 derive the full matrix  $\text{Var}(\underline{\hat{\beta}}|X's)$ . *Hint: make use of the expression for  $\text{Var}(\hat{\beta}_1|X's)$  and the result from 2.f above.*
4. Suppose that we have three independent observations  $(X_i, Y_i)$  of the random vector  $(X, Y)$ . Assume that  $E(Y_i|X_i) = 2X_i + 3$ , and  $\text{Var}(Y_i|X_i) = 3$ .
- (a) If  $X_1 = 1$ ,  $X_2 = 3$ , and  $X_3 = 6.5$ , is assumption 3 fulfilled?
  - (b) Find the numeric values of  $E(\bar{Y} - \hat{\beta}_1 X_1|X's)$ , and  $\text{Var}(\hat{\beta}_0 + \hat{\beta}_1 X_1|X's)$ .
  - (c) Indicate the numeric values of  $\text{Var}(\underline{\hat{\beta}}|X's)$  (the whole matrix).
5. Suppose that we have  $n$  independent observations  $(X_i, Y_i)$  of the random vector  $(X, Y)$  with pdf  $f(x, y)$ . Show that
- (a)  $f(y_1, \dots, y_n|x_1, \dots, x_n) = \prod_{i=1}^n f(y_i|x_i)$ . This is called *conditional independence*.
  - (b)  $f(y_i|x_1, \dots, x_n) = f(y_i|x_i)$  for  $i = 1, \dots, n$ .
  - (c)  $f(y_i, y_j|x_1, \dots, x_n) = f(y_i|x_i)f(y_j|x_j)$  for any  $i \neq j$ .
  - (d) If  $\text{Var}(Y_1) < \infty$  then  $\text{Cov}(Y_i, Y_j|X_1, \dots, X_n) = 1_{\{0\}}(i - j)\text{Var}(Y_1|X_1)$ .
6. Suppose that we have  $n$  independent observations  $(X_i, Y_i)$  of the random vector  $(X, Y)$  with pdf  $f(x, y) = (x + y)1_{[0,1]}(x)1_{[0,1]}(y)$ . Let  $(\hat{\beta}_0, \hat{\beta}_1)$  be the LSE. Find  $E(\hat{\beta}_1|X's)$  and compute its numerical value when  $X_1 = 1$ ,  $X_2 = 3$ , and  $X_3 = 6.5$ .
7. By definition, a Student  $t$  random variable with  $k$  degrees of freedom has the form

$$t = \frac{Z}{\sqrt{V/k}}$$

where  $Z \sim N(0, 1)$ ,  $V \sim \chi^2(k)$  and  $Z$  and  $V$  are independent. Assuming that  $(\hat{\beta}_0, \hat{\beta}_1)$  and  $\hat{\sigma}^2$  are independent, show that

$$\frac{\hat{\beta}_i - \beta_i}{\text{SE}(\hat{\beta}_i|X's)} | X's \sim t(n-2)$$

8. Use the result from problem 7 to construct a  $(1 - \alpha)\%$  confidence interval for  $\beta_1$ . Find the numerical value of the interval if  $\alpha = 0.05$ ,  $n = 3$ ,  $X_1 = 1$ ,  $X_2 = 3$ ,  $X_3 = 6.5$ , and  $\sum(\hat{\epsilon}_i)^2 = 8.6$ .